



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

• Experiment No.: 1

1. Experiment Title:

Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

2. Software/Tools Required:

Visual Studio Code, Web Browser (Google Chrome / Mozilla Firefox), Node.js (for local server execution)

3. Experiment Program Code:

- expl_inline.html Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Hello World.</title>
</head>
<body style="background-color:#f0f8ff; text-align:center; font-
family:Arial, sans-serif; padding:40px;">

  <h1 style="font-size:48px; color:white; background-color:#007BFF;
padding:15px; border-radius:8px;">
    Hello World
  </h1>
<p>
  Name : Manaswi Sakharwade , PRN : 24070521169
</p>
  <p style="font-size:20px; color:#333333; background-color:#ffffcc;
padding:10px; margin-top:20px;">

    This is a sample webpage designed by Inline CSS.
  </p>

  <button style="font-size:18px; color:white; background-color:green;
padding:12px 24px; border:none; border-radius:5px; cursor:pointer;
margin-top:20px;">
    Click Me
  </button>
```

```
</body>
</html>
```

- expl_external.html Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Hello World</title>

  <style>
    body{
      background-color:#ffe4e1;
      text-align:center;
      font-family:Arial, sans-serif;
      padding:40px;
    }

    h1{
      font-size:48px;
      color:white;
      background-color:#91ff00;
      padding:15px;
      border-radius:8px;
    }

    p{
      font-size:20px;
      color:#333333;
      background-color:#ffffcc;
      padding:10px;
      margin-top:20px;
    }

    button{
      font-size:18px;
      color:white;
      background-color:green;
      padding:12px 24px;
      border:none;
      border-radius:5px;
      cursor:pointer;
      margin-top:20px;
    }
  </style>

</head>
<body>
```

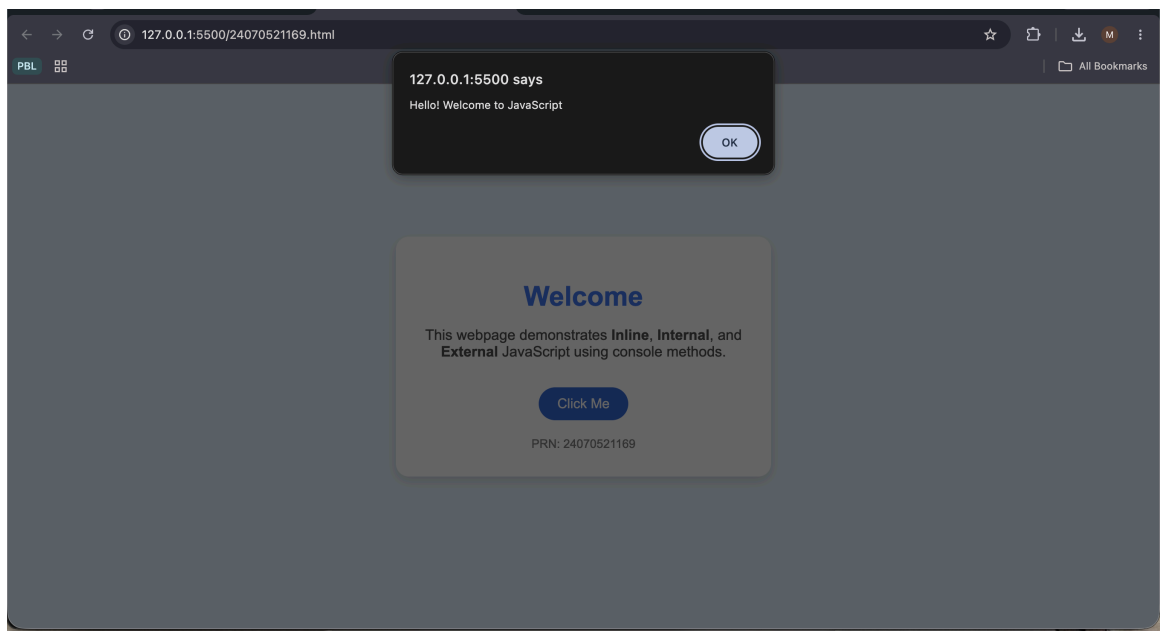
```
<h1>Hello World</h1>
<p>Name : Manaswi Sakharwade , PRN : 24070521169</p>

<p>This is a sample Webpage made using External css.</p>

<button>Click Me</button>

</body>
</html>
```

4. Output:



5. Case Study 01

Case Study Title:

Create a web page for college department and link the student information page with it and write a js program to display "Welcome to SIT Nagpur" in an alert box

Case Study Code:

- expl_casestudy.html Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
```

```

<title>Department of CSE - SIT Nagpur</title>
<style>
  :root {
    --primary: #0f172a;
    --accent: #2563eb;
    --accent-hover: #1d4ed8;
    --bg: #f8fafc;
    --card-bg: #ffffff;
    --text-main: #334155;
    --text-dark: #0f172a;
  }

  body {
    font-family: 'Segoe UI', system-ui, -apple-system, sans-
serif;
    background-color: var(--bg);
    color: var(--text-main);
    margin: 0;
    padding: 0;
    line-height: 1.6;
  }

  /* Top Navbar */
  .navbar {
    background-color: var(--primary);
    color: white;
    padding: 15px 5%;
    display: flex;
    justify-content: space-between;
    align-items: center;
    box-shadow: 0 4px 6px -1px rgba(0,0,0,0.1);
  }

  .logo {
    font-weight: 700;
    font-size: 1.4rem;
    letter-spacing: -0.5px;
  }

  /* Hero Banner Section */
  .hero {
    background: linear-gradient(135deg, #1e3a8a 0%, #0f172a
100%);
    color: white;
    padding: 60px 20px;
    text-align: center;
  }

  .hero h1 {
    font-size: 2.5rem;
    margin: 0 0 10px 0;
  }

```

```

        font-weight: 800;
    }

    .hero p {
        font-size: 1.1rem;
        opacity: 0.9;
        max-width: 600px;
        margin: 0 auto;
    }

    /* Main Container */
    .main-container {
        max-width: 1100px;
        margin: -40px auto 40px auto;
        padding: 0 20px;
    }

    /* Stats Grid */
    .stats-grid {
        display: grid;
        grid-template-columns: repeat(auto-fit, minmax(220px,
1fr));
        gap: 20px;
        margin-bottom: 40px;
    }

    .stat-card {
        background: var(--card-bg);
        padding: 25px;
        border-radius: 12px;
        text-align: center;
        box-shadow: 0 10px 15px -3px rgba(0,0,0,0.05);
        border: 1px solid #e2e8f0;
        transition: transform 0.2s;
    }

    .stat-card:hover {
        transform: translateY(-5px);
    }

    .stat-number {
        font-size: 2rem;
        font-weight: 700;
        color: var(--accent);
        margin-bottom: 5px;
    }

    .stat-label {
        font-size: 0.9rem;
        font-weight: 600;
        color: var(--text-dark);
    }

```

```

    }

    /* Content Card */
    .content-card {
        background: var(--card-bg);
        padding: 40px;
        border-radius: 16px;
        box-shadow: 0 10px 15px -3px rgba(0,0,0,0.05);
        border: 1px solid #e2e8f0;
        text-align: center;
    }

    .content-card h2 {
        color: var(--text-dark);
        font-size: 1.8rem;
        margin-top: 0;
    }

    /* Call to Action Button */
    .cta-btn {
        display: inline-flex;
        align-items: center;
        background-color: var(--accent);
        color: white;
        text-decoration: none;
        padding: 14px 32px;
        border-radius: 8px;
        font-weight: 600;
        font-size: 1rem;
        margin-top: 25px;
        box-shadow: 0 4px 14px rgba(37, 99, 235, 0.3);
        transition: all 0.2s ease;
    }

    .cta-btn:hover {
        background-color: var(--accent-hover);
        transform: translateY(-2px);
        box-shadow: 0 6px 20px rgba(37, 99, 235, 0.4);
    }

    .cta-btn span {
        margin-left: 8px;
        transition: transform 0.2s;
    }

    .cta-btn:hover span {
        transform: translateX(4px);
    }
</style>
</head>
<body onload="showAlert()">

```

```

<!-- Navbar -->
<nav class="navbar">
    <div class="logo">SIT NAGPUR</div>
    <div>Academic Year 2026-27</div>
</nav>

<!-- Hero Area -->
<header class="hero">
    <h1>Welcome to SIT Nagpur</h1>
    <p>Shaping the future through innovation, research, and
excellent engineering education.</p>
</header>

<!-- Core Content Container -->
<main class="main-container">

    <!-- Quick Stats Overview -->
    <section class="stats-grid">
        <div class="stat-card">
            <div class="stat-number">1200+</div>
            <div class="stat-label">Active Students</div>
        </div>
        <div class="stat-card">
            <div class="stat-number">45+</div>
            <div class="stat-label">Expert Faculty</div>
        </div>
        <div class="stat-card">
            <div class="stat-number">94%</div>
            <div class="stat-label">Placement Record</div>
        </div>
    </section>

    <!-- Department Hub -->
    <section class="content-card">
        <h2>Computer Science & Engineering Portal</h2>
        <p>Access your curriculum tracking tools, manage course
registration profiles, review upcoming exam timetables, and securely
update your academic portfolios from our unified digital hub.</p>

        <a href="student_info.html" class="cta-btn">
            Access Student Information Page <span>&rarr;</span>
        </a>
    </section>
</main>

<!-- JavaScript Alert Script -->
<script>
    function showAlert() {
        alert("Welcome to SIT NAGPUR");
    }

```

```
</script>

</body>
</html>
```

- student.html Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Student Profile - SIT Nagpur</title>
  <style>
    :root {
      --primary: #0f172a;
      --accent: #2563eb;
      --bg: #f8fafc;
      --card-bg: #ffffff;
      --text-main: #334155;
      --text-dark: #0f172a;
    }

    body {
      font-family: 'Segoe UI', system-ui, -apple-system, sans-
serif;
      background-color: var(--bg);
      color: var(--text-main);
      margin: 0;
      padding: 0;
      line-height: 1.6;
    }

    .navbar {
      background-color: var(--primary);
      color: white;
      padding: 15px 5%;
      display: flex;
      justify-content: space-between;
      align-items: center;
    }

    .logo {
      font-weight: 700;
      font-size: 1.4rem;
    }

    .container {
      max-width: 800px;
      margin: 40px auto;
```

```

        padding: 0 20px;
    }

    .profile-card {
        background: var(--card-bg);
        border-radius: 16px;
        box-shadow: 0 10px 15px -3px rgba(0,0,0,0.05);
        border: 1px solid #e2e8f0;
        padding: 40px;
    }

    h1 {
        color: var(--text-dark);
        margin-top: 0;
        font-size: 2rem;
        border-bottom: 2px solid #e2e8f0;
        padding-bottom: 15px;
    }

    /* Modern Table Styling */
    .info-table {
        width: 100%;
        border-collapse: collapse;
        margin-top: 25px;
    }

    .info-table th, .info-table td {
        padding: 14px 16px;
        text-align: left;
        border-bottom: 1px solid #e2e8f0;
    }

    .info-table th {
        background-color: #f1f5f9;
        color: var(--text-dark);
        font-weight: 600;
        width: 30%;
    }

    .back-link {
        display: inline-block;
        margin-top: 30px;
        color: var(--accent);
        text-decoration: none;
        font-weight: 600;
        transition: transform 0.2s;
    }

    .back-link:hover {
        text-decoration: underline;
        transform: translateX(-3px);
    }

```

```

    }
  </style>
</head>
<body>

  <!-- Navbar -->
  <nav class="navbar">
    <div class="logo">SIT NAGPUR</div>
    <div>Dashboard</div>
  </nav>

  <!-- Main Container -->
  <div class="container">
    <div class="profile-card">
      <h1>Student Information Profile</h1>
      <p>Verify your enrollment status and core registration
details below. Contact the department office to modify locked fields.</
p>

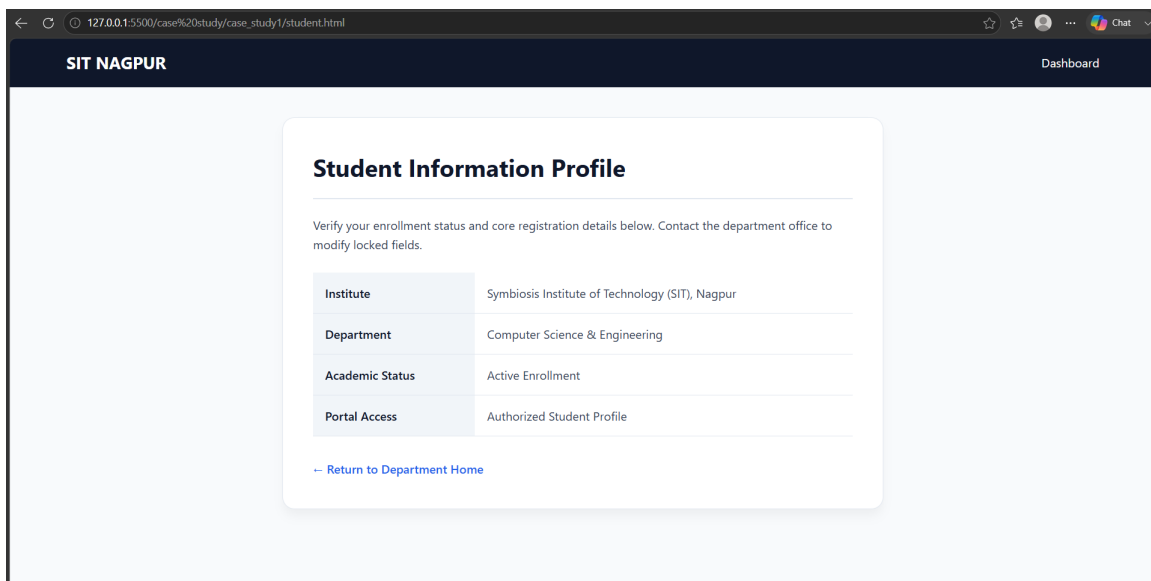
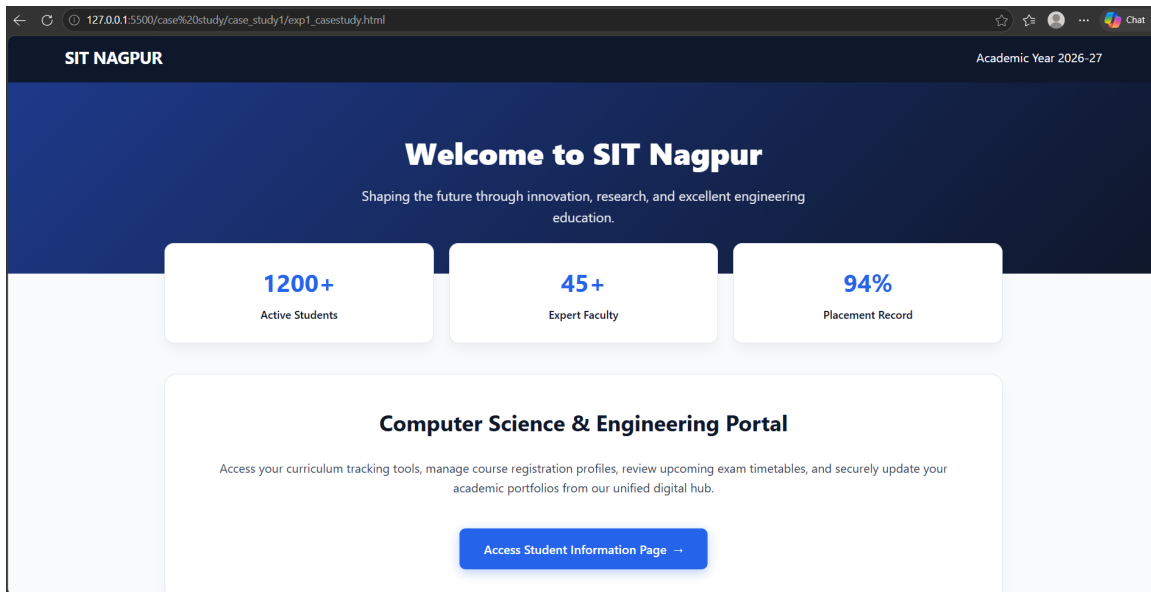
      <!-- Sample Data Layout -->
      <table class="info-table">
        <tr>
          <th>Institute</th>
          <td>Symbiosis Institute of Technology (SIT),
Nagpur</td>
        </tr>
        <tr>
          <th>Department</th>
          <td>Computer Science & Engineering</td>
        </tr>
        <tr>
          <th>Academic Status</th>
          <td>Active Enrollment</td>
        </tr>
        <tr>
          <th>Portal Access</th>
          <td>Authorized Student Profile</td>
        </tr>
      </table>

      <a class="back-link" href="index.html">&larr; Return to
Department Home</a>
    </div>
  </div>

</body>
</html>

```

Case Study Output:



Result/Conclusion:

The experiment was successfully executed. The webpage correctly demonstrated the use of inline, internal, and external JavaScript to trigger different interactive elements. Furthermore, the user information was dynamically rendered onto the page, and the various console methods were successfully utilized to log data and warnings in the browser's developer tools.



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Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

• Experiment No.: 2

1. Experiment Title:

Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

2. Software/Tools Required:

Visual Studio Code, Web Browser (Google Chrome / Mozilla Firefox), Node.js (for local server execution)

3. Experiment Program Code:

- exp2.html Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Bill Receipt</title>
  <style>
    body {
      background-color: #fafc9a;
      font-family: Arial, sans-serif;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      margin: 0;
    }
    .receipt-box {
      background-color: #9d9e3f;
      border: 2px solid #333333;
      padding: 30px;
      width: 350px;
      box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
    }
    h3 {
      text-align: center;
      margin-top: 0;
    }
  </style>
```

```

</head>
<body>

    <script>
        // 1. Popups to take user input
        var uName = prompt("Enter User Name:");
        var contact = prompt("Enter Contact No:");
        var pName = prompt("Enter Product Name:");
        var qty = Number(prompt("Enter Quantity:"));
        var price = Number(prompt("Enter Price per Unit:"));

        // 2. Mathematical calculations
        var total = qty * price;
        var gst = total * 0.18;
        var finalAmount = total + gst;

        // 3. Outputting only the centered bill box
        document.write("<div class='receipt-box'>");
        document.write("<h3>Bill Receipt</h3>");
        document.write("<p><b>User Name:</b> " + uName + "</p>");
        document.write("<p><b>Contact No:</b> " + contact + "</p>");
        document.write("<p><b>Product Name:</b> " + pName + "</p>");
        document.write("<p><b>Quantity:</b> " + qty + "</p>");
        document.write("<p><b>Price:</b> " + price + "</p>");
        document.write("<hr>");
        document.write("<p><b>Total Value:</b> " + total + "</p>");
        document.write("<p><b>GST (18%):</b> " + gst + "</p>");
        document.write("<p style='font-size: 18px;'><b>Final Amount:</b>");
    b> " + finalAmount + "</p>");
        document.write("<br>");
        document.write("<p style='font-size: 12px; color: #555; text-align: center;'>Name: Manaswi Sakharwade , PRN: 24070521169</p>");
        document.write("</div>");
    </script>

</body>
</html>

```

4. Output:

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5500/24070521169_Calculator.html". The page content is a "Billing Calculator" form. The form has three input fields: "Name" (containing "Manaswi Sakharwade"), "Contact No" (containing "6"), and "Quantity" (containing "7"). Below these fields is a blue "Calculate Bill" button. Under the button, the results are displayed in a light blue box with a white background. The results are: "ABC Store", "Customer: Manaswi Sakharwade", "Subtotal: ₹42.00", "GST (18%): ₹7.56", "Total Bill: ₹49.56", and "PRN: 24070521169".

5. Case Study 02

Case Study Title:

Online Shopping - Bill Calculator

Case Study Code:

- case_study2.html Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Online Shopping & Billing Portal</title>
  <style>
    :root {
      --primary-color: #007bff;
      --success-color: #28a745;
      --bg-color: #f8f9fa;
      --card-bg: #ffffff;
      --text-color: #333333;
    }

    body {
      font-family: Arial, sans-serif;
      background-color: var(--bg-color);
      color: var(--text-color);
      margin: 0;
      padding: 20px;
    }

    .container {
      max-width: 800px;
      margin: 0 auto;
      background: var(--card-bg);
      padding: 30px;
      border-radius: 8px;
      box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
    }

    h1 {
      text-align: center;
      color: var(--primary-color);
      margin-bottom: 25px;
    }

    fieldset {
      border: 1px solid #ced4da;
      border-radius: 6px;
```

```
padding: 20px;
margin-bottom: 20px;
}

legend {
  font-weight: bold;
  color: var(--primary-color);
  padding: 0 10px;
}

.form-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 15px;
}

.form-group {
  display: flex;
  flex-direction: column;
}

.form-group.full-width {
  grid-column: span 2;
}

label {
  font-size: 14px;
  font-weight: bold;
  margin-bottom: 5px;
}

input, select {
  padding: 10px;
  border: 1px solid #ccc;
  border-radius: 4px;
  font-size: 14px;
}

button {
  background-color: var(--success-color);
  color: white;
  padding: 12px;
  border: none;
  border-radius: 4px;
  font-size: 16px;
  font-weight: bold;
  cursor: pointer;
  width: 100%;
  margin-top: 10px;
}
```

```

button:hover {
    opacity: 0.9;
}

.invoice-output {
    margin-top: 30px;
    background: #eef2f7;
    padding: 20px;
    border-radius: 6px;
    border-left: 5px solid var(--primary-color);
}

.invoice-output h2 {
    margin-top: 0;
    color: #2c3e50;
}

.bill-row {
    display: flex;
    justify-content: space-between;
    margin-bottom: 8px;
    font-size: 15px;
}

.bill-row.total {
    font-weight: bold;
    font-size: 18px;
    border-top: 1px solid #ccc;
    padding-top: 10px;
    margin-top: 10px;
    color: var(--success-color);
}
</style>
</head>
<body>

<div class="container">
    <h1>Online Shopping Store</h1>

    <form id="billingForm" onsubmit="calculateBill(event)">
        <!-- Customer Details -->
        <fieldset>
            <legend>Customer Information</legend>
            <div class="form-grid">
                <div class="form-group">
                    <label for="custName">Customer Name:</label>
                    <input type="text" id="custName" required
placeholder="Enter full name">
                </div>
                <div class="form-group">
                    <label for="mobileNo">Mobile Number:</label>

```

```

        <input type="tel" id="mobileNo" pattern="[0-9]{10}"
required placeholder="10-digit mobile number">
    </div>
    <div class="form-group">
        <label for="invoiceNo">Invoice Number:</label>
        <input type="text" id="invoiceNo"
value="INV-2026-001" readonly>
    </div>
    <div class="form-group">
        <label for="membership">Membership Type:</label>
        <select id="membership">
            <option value="0">None (0% Extra Off)</option>
            <option value="5">Silver (5% Extra Off)</
option>
            <option value="10">Gold (10% Extra Off)</
option>
        </select>
    </div>
</div>
</fieldset>

<!-- Product & Purchase Details -->
<fieldset>
    <legend>Product & Charges Details</legend>
    <div class="form-grid">
        <div class="form-group">
            <label for="productName">Product Name:</label>
            <input type="text" id="productName" required
placeholder="e.g. Organic Cashews">
        </div>
        <div class="form-group">
            <label for="quantityKg">Quantity (in kg):</label>
            <input type="number" id="quantityKg" step="0.1"
min="0.1" required placeholder="e.g. 2.5">
        </div>
        <div class="form-group">
            <label for="rateKg">Rate per kg (₹):</label>
            <input type="number" id="rateKg" step="1" min="1"
required placeholder="e.g. 500">
        </div>
        <div class="form-group">
            <label for="gstRate">GST Rate (%):</label>
            <select id="gstRate">
                <option value="5">5%</option>
                <option value="12">12%</option>
                <option value="18">18%</option>
            </select>
        </div>
        <div class="form-group">
            <label for="packingCharges">Packing Charges (₹):</
label>

```

```

        <input type="number" id="packingCharges" value="30"
min="0">
        </div>
        <div class="form-group">
            <label for="paymentMode">Payment Mode:</label>
            <select id="paymentMode">
                <option value="UPI">UPI</option>
                <option value="Credit Card">Credit Card</
option>
                <option value="Debit Card">Debit Card</option>
                <option value="Cash on Delivery">Cash on
Delivery</option>
            </select>
        </div>
    </div>
</fieldset>

    <button type="submit">Calculate and Generate Bill</button>
</form>

    <!-- Computed Invoice Display Area -->
    <div class="invoice-output" id="invoiceSection" style="display:
none;">
        <h2>Invoice Summary</h2>
        <div class="bill-row"><span>Invoice No:</span> <span
id="outInv"></span></div>
        <div class="bill-row"><span>Customer:</span> <span
id="outName"></span></div>
        <div class="bill-row"><span>Mobile:</span> <span
id="outMobile"></span></div>
        <div class="bill-row"><span>Product:</span> <span
id="outProduct"></span></div>
        <div class="bill-row"><span>Qty & Rate:</span> <span
id="outQtyRate"></span></div>
        <div class="bill-row"><span>Base Amount:</span> <span
id="outBase"></span></div>
        <div class="bill-row"><span>Membership Disc:</span> <span
id="outMemberDisc"></span></div>
        <div class="bill-row"><span>GST Amount:</span> <span
id="outGst"></span></div>
        <div class="bill-row"><span>Packing Charges:</span> <span
id="outPacking"></span></div>
        <div class="bill-row"><span>Payment Mode:</span> <span
id="outPayMode"></span></div>
        <div class="bill-row total"><span>Final Payable Total:</span>
<span id="outTotal"></span></div>
    </div>
</div>

<script>
    function calculateBill(event) {

```

```

    event.preventDefault();

    // Fetch inputs
    const name = document.getElementById('custName').value;
    const mobile = document.getElementById('mobileNo').value;
    const inv = document.getElementById('invoiceNo').value;
    const product = document.getElementById('productName').value;
    const qty =
parseFloat(document.getElementById('quantityKg').value);
    const rate =
parseFloat(document.getElementById('rateKg').value);
    const memberDiscPercent =
parseFloat(document.getElementById('membership').value);
    const gstPercent =
parseFloat(document.getElementById('gstRate').value);
    const packing =
parseFloat(document.getElementById('packingCharges').value);
    const payMode = document.getElementById('paymentMode').value;

    // Calculations
    const baseAmount = qty * rate;
    const discountAmount = (baseAmount * memberDiscPercent) / 100;
    const amountAfterDiscount = baseAmount - discountAmount;
    const gstAmount = (amountAfterDiscount * gstPercent) / 100;
    const finalTotal = amountAfterDiscount + gstAmount + packing;

    // Bind outputs
    document.getElementById('outInv').innerText = inv;
    document.getElementById('outName').innerText = name;
    document.getElementById('outMobile').innerText = mobile;
    document.getElementById('outProduct').innerText = product;
    document.getElementById('outQtyRate').innerText = qty + ' kg @
₹' + rate + '/kg';
    document.getElementById('outBase').innerText = '₹' +
baseAmount.toFixed(2);
    document.getElementById('outMemberDisc').innerText = '- ₹' +
discountAmount.toFixed(2) + ' (' + memberDiscPercent + '%)';
    document.getElementById('outGst').innerText = '+ ₹' +
gstAmount.toFixed(2) + ' (' + gstPercent + '%)';
    document.getElementById('outPacking').innerText = '+ ₹' +
packing.toFixed(2);
    document.getElementById('outPayMode').innerText = payMode;
    document.getElementById('outTotal').innerText = '₹' +
finalTotal.toFixed(2);

    // Show invoice section
    document.getElementById('invoiceSection').style.display =
'block';
}
</script>

```

```
</body>
</html>
```

Case Study Output:

The screenshot displays a web browser window with the address bar showing '127.0.0.1:5500/case2.html'. The page title is 'Online Shopping Store'. The interface is divided into three main sections: Customer Details, Product Details, and Invoice Summary.

Customer Details:

- Name: Manaswi Sakharwade
- Mobile Number: 8589472575
- Invoice Number: INV-2026-001
- Membership: None

Product Details:

- Product Name: cap
- Quantity (kg): 86
- Rate per kg (₹): 76
- GST: 5%
- Packing Charges (₹): 30
- Payment Mode: UPI

Generate Bill

Invoice Summary:

Invoice No.	INV-2026-001
Customer	Manaswi Sakharwade
Mobile	8589472575
Product	cap
Quantity & Rate	86 kg @ ₹76
Base Amount	₹6536.00
Membership Discount	- ₹0.00
GST	+ ₹326.80
Packing Charges	₹30.00
Payment Mode	UPI
Final Total	₹6892.80

Name: Manaswi Sakharwade
PRN: 24070521169

Result/Conclusion:

The billing calculator was successfully built. The program accurately captured user input for item details, calculated the subtotal and tax, and applied destructuring to efficiently extract the calculated values. Template literals were effectively utilized to dynamically generate and display a formatted receipt on the webpage.



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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

• Experiment No.: 3

1. Experiment Title:

Implement control structures and form validation; create a grading system based on user-entered marks.

2. Software/Tools Required:

Visual Studio Code, Web Browser (Google Chrome / Mozilla Firefox), Node.js (for local server execution)

3. Experiment Program Code:

- exp3.html Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>5-Subject Grading System</title>
  <style>
    * { box-sizing: border-box; font-family: system-ui, -apple-system,
sans-serif; }
    body { background-color: #b3ebb3; display: flex; justify-content:
center; align-items: center; min-height: 100vh; margin: 0; padding:
1.5rem; }
    .card { background: #e3d9f3; padding: 2rem; border-radius: 12px;
box-shadow: 0 4px 16px rgba(0,0,0,0.08); width: 100%; max-width: 480px;
}
    h2 { margin-top: 0; color: #1e293b; text-align: center; }
    .grid { display: grid; grid-template-columns: 1fr 1fr; gap: 0.75rem
1rem; }
    .form-group { margin-bottom: 0.75rem; }
    .full-width { grid-column: span 2; }
    label { display: block; margin-bottom: 0.35rem; font-weight: 600;
color: #334155; font-size: 0.9rem; }
    input[type="number"] { width: 100%; padding: 0.65rem; border: 1px
solid #cbd5e1; border-radius: 6px; font-size: 0.95rem; }
    input[type="number"]:focus { outline: none; border-color: #2563eb;
```

```

box-shadow: 0 0 0 3px rgba(37,99,235,0.15); }
    button { width: 100%; padding: 0.75rem; background-color: #2563eb;
color: white; border: none; border-radius: 6px; font-size: 1rem; font-
weight: 600; cursor: pointer; margin-top: 0.5rem; transition:
background 0.2s; }
    button:hover { background-color: #1d4ed8; }
    .error-banner { background-color: #fef2f2; border-left: 4px solid
#ef4444; color: #991b1b; padding: 0.75rem; border-radius: 4px; font-
size: 0.875rem; margin-bottom: 1rem; display: none; }
    .result-box { margin-top: 1.25rem; padding: 1.25rem; border-radius:
8px; display: none; }
    .result-header { font-size: 1.1rem; font-weight: 700; margin-
bottom: 0.5rem; text-align: center; }
    .result-details { font-size: 0.95rem; line-height: 1.5; }
    /* Dynamic Theme Colors */
    .grade-A { background-color: #dcfce7; color: #166534; border: 1px
solid #bbf7d0; }
    .grade-B { background-color: #e0f2fe; color: #075985; border: 1px
solid #bae6fd; }
    .grade-C { background-color: #fef9c3; color: #854d0e; border: 1px
solid #fef08a; }
    .grade-D { background-color: #ffedd5; color: #9a3412; border: 1px
solid #fed7aa; }
    .grade-F { background-color: #fee2e2; color: #991b1b; border: 1px
solid #fca5a5; }
    </style>
</head>
<body>

    <div class="card">
        <h2>Student Assessment</h2>

        <!-- Unified Error Banner -->
        <div id="errorBanner" class="error-banner"></div>

        <form id="gradeForm" novalidate>
            <div class="grid">
                <div class="form-group">
                    <label for="sub1">Mathematics</label>
                    <input type="number" id="sub1" class="subject-input"
placeholder="0 - 100">
                </div>
                <div class="form-group">
                    <label for="sub2">Science</label>
                    <input type="number" id="sub2" class="subject-input"
placeholder="0 - 100">
                </div>
                <div class="form-group">
                    <label for="sub3">English</label>
                    <input type="number" id="sub3" class="subject-input"
placeholder="0 - 100">

```

```

    </div>
    <div class="form-group">
      <label for="sub4">History</label>
      <input type="number" id="sub4" class="subject-input"
placeholder="0 - 100">
    </div>
    <div class="form-group full-width">
      <label for="sub5">Computer Science</label>
      <input type="number" id="sub5" class="subject-input"
placeholder="0 - 100">
    </div>
  </div>
  <button type="submit">Calculate Final Results</button>
</form>

<div id="resultBox" class="result-box">
  <div id="resultHeader" class="result-header"></div>
  <div id="resultDetails" class="result-details"></div>
</div>
</div>

<script>
  document.getElementById('gradeForm').addEventListener('submit',
function(e) {
    e.preventDefault();

    const inputs = document.querySelectorAll('.subject-input');
    const errorBanner = document.getElementById('errorBanner');
    const resultBox = document.getElementById('resultBox');
    const resultHeader = document.getElementById('resultHeader');
    const resultDetails = document.getElementById('resultDetails');

    // Reset UI state
    errorBanner.style.display = 'none';
    resultBox.style.display = 'none';
    resultBox.className = 'result-box';

    const scores = [];
    let hasFailedSubject = false;

    // --- 1. CONTROL STRUCTURE: Loop with Validation Checks ---
    for (let i = 0; i < inputs.length; i++) {
      const value = inputs[i].value.trim();
      const label = inputs[i].previousElementSibling.textContent;

      // Check empty input
      if (value === '') {
        showError(`Please fill in marks for <strong>${label}</
strong>.`);
        inputs[i].focus();
        return;

```

```

    }

    const score = Number(value);

    // Check invalid numbers or out-of-bounds inputs
    if (isNaN(score) || score < 0 || score > 100) {
        showError(`Invalid mark for <strong>${label}</strong>. Enter
a value between 0 and 100.`);
        inputs[i].focus();
        return;
    }

    // Check for subject failure condition (e.g., passing mark is
40)    if (score < 40) {
        hasFailedSubject = true;
    }

    scores.push(score);
}

// --- 2. CALCULATIONS ---
const totalMarks = scores.reduce((sum, current) => sum + current,
0);

const percentage = (totalMarks / (scores.length * 100)) * 100;

// --- 3. CONTROL STRUCTURE: Grading Logic ---
let grade = '';
let status = '';
let cssClass = '';

if (hasFailedSubject) {
    // Essential academic condition: Failing any individual subject
    overrides average grade
    status = 'FAILED (Failed 1 or more subjects)';
    grade = 'F';
    cssClass = 'grade-F';
} else if (percentage >= 90) {
    status = 'PASSED';
    grade = 'A+ (Outstanding)';
    cssClass = 'grade-A';
} else if (percentage >= 80) {
    status = 'PASSED';
    grade = 'A (Excellent)';
    cssClass = 'grade-A';
} else if (percentage >= 70) {
    status = 'PASSED';
    grade = 'B (Good)';
    cssClass = 'grade-B';
} else if (percentage >= 60) {
    status = 'PASSED';

```

```

        grade = 'C (Satisfactory)';
        cssClass = 'grade-C';
    } else if (percentage >= 50) {
        status = 'PASSED';
        grade = 'D (Pass)';
        cssClass = 'grade-D';
    } else {
        status = 'FAILED';
        grade = 'F (Needs Improvement)';
        cssClass = 'grade-F';
    }

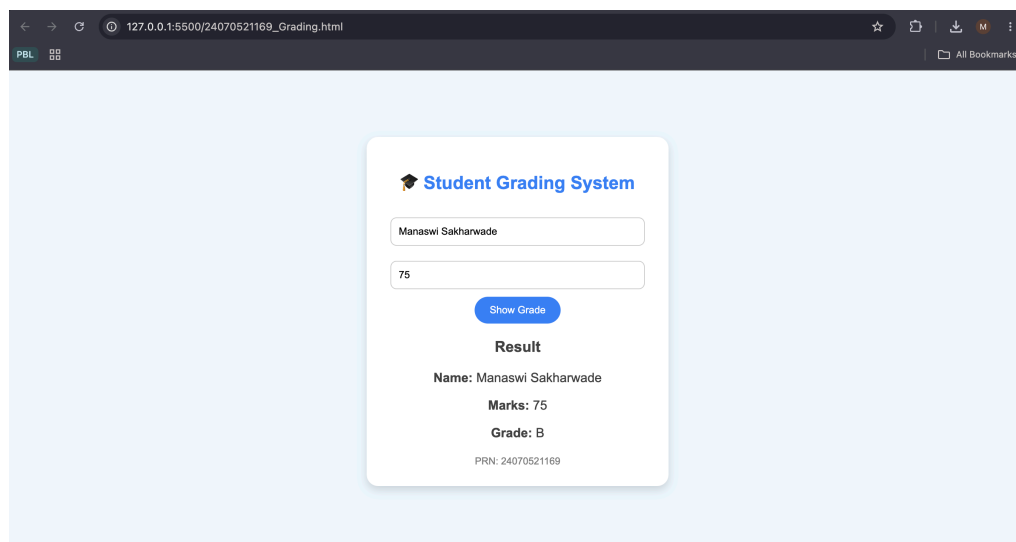
    // --- 4. RENDER OUTPUT ---
    resultHeader.textContent = `Status: ${status}`;
    resultDetails.innerHTML = `
        <strong>Total Marks:</strong> ${totalMarks} / 500<br>
        <strong>Overall Percentage:</strong> ${percentage.toFixed(2)}
    %<br>
        <strong>Final Grade:</strong> ${grade}<br>
        <hr style="margin: 0.75rem 0; border: none; border-top: 1px
dashed rgba(0,0,0,0.2);">
        <strong>Student Name:</strong> Manaswi Sakharwade<br>
        <strong>PRN:</strong> 24070521169
    `;

    resultBox.classList.add(cssClass);
    resultBox.style.display = 'block';
});

function showError(htmlMessage) {
    const errorBanner = document.getElementById('errorBanner');
    errorBanner.innerHTML = htmlMessage;
    errorBanner.style.display = 'block';
}
</script>
</body>
</html>

```

4. Output:



5. Case Study 03

Case Study Title:

Scholarship Eligibility System

Case Study Code:

- case_study3.html Code:

```
<html>
  <head>
    <title>Scholarship Eligibility</title>
    <style>
      body {
        font-family: Arial, sans-serif;
        background-color: #f2f2f2;
        text-align: center;
        padding: 40px;
      }
      h2 {
        color: #c21321af;
        margin: 5px 0;
      }
      input {
        padding: 8px;
        font-size: 15px;
        margin-top: 10px;
        border: 1px solid #c0e61b;
        border-radius: 4px;
      }
      button {
        display: block;
        margin: 15px auto 0;
        padding: 8px 20px;
        font-size: 15px;
        background-color: #1a5cff;
        color: white;
        border: none;
        border-radius: 4px;
        cursor: pointer;
      }
      button:hover {
        background-color: #0d3fb0;
      }
      #result {
        margin-top: 20px;
        font-size: 17px;
        color: #333;
      }
    </style>
  </head>
```

```

<body>
  <h2>Manaswi Sakharwade 24070521169</h2>
  <h2>Scholarship Eligibility System</h2>
  <h3>Enter marks:</h3>
  <input type='number' id='marks'></input>
  <button onClick="checkScholarship()">Submit</button>
  <p id='result'></p>
  <script>
    function checkScholarship(){
      var marks = document.getElementById("marks").value;
      if( marks == ""){
        alert("Please Enter marks!!");
        return;
      }
      if(marks<0 || marks> 100){
        alert("Enter marks between 0 and 100");
        return;
      }
      if(marks>=95){
        scholarship = "Platinum Scholarship";
      }
      else if(marks>=85){
        scholarship = "Gold Scholarship";
      }
      else if(marks>=70){
        scholarship = "Silver Scholarship";
      }
      else if(marks>=50){
        scholarship = "Bronze Scholarship";
      }
      else{
        scholarship = "Not Eligible";
      }
      document.getElementById("result").innerHTML =
        "Marks = " + marks + "<br>Scholarship = " +
scholarship;
    }
  </script>
</body>
</html>

```

Case Study Output:

Manaswi Sakharwade
PRN: 24070521169
Scholarship Eligibility System

Enter Marks

64

Submit

Student Name: Manaswi Sakharwade
PRN: 24070521169

Marks = 64
Scholarship = Bronze Scholarship

Result/Conclusion:

The grading system was successfully implemented. The application correctly validated the user's input, ensuring that only numerical marks between 0 and 100 were accepted. The control structures functioned as expected, accurately assigning the correct grade and pass/fail status based on the defined mark boundaries.



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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

• Experiment No.: 4

1. Experiment Title:

Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. Software/Tools Required:

Visual Studio Code, Web Browser (Google Chrome / Mozilla Firefox), Node.js (for local server execution)

3. Experiment Program Code:

- exp4.html Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Palindrome Checker</title>

  <style>
    *{
      margin:0;
      padding:0;
      box-sizing:border-box;
      font-family:Arial, sans-serif;
    }

    body{
      background-color:#EAF4F4; /* Subtle light blue-green
background */
      display:flex;
      justify-content:center;
      align-items:center;
      height:100vh;
    }

    .container{
```

```
        width:380px;
        background:#ffffff;
        padding:25px;
        border-radius:12px;
        text-align:center;
        box-shadow:0 8px 20px rgba(0,0,0,0.15);
    }

    h2{
        margin-bottom:20px;
        color:#2C3E50;
    }

    input{
        width:100%;
        padding:10px;
        border:1px solid #bfc9ca;
        border-radius:6px;
        font-size:16px;
        outline:none;
        margin-bottom:15px;
    }

    input:focus{
        border-color:#5DADE2;
    }

    button{
        background:#5DADE2;
        color:white;
        border:none;
        padding:10px 20px;
        border-radius:6px;
        cursor:pointer;
        font-size:16px;
    }

    button:hover{
        background:#3498DB;
    }

    #result{
        margin-top:18px;
        font-size:18px;
        font-weight:bold;
        color:#2E4053;
    }

    .footer{
        margin-top:25px;
        padding-top:15px;
    }
```

```

        border-top:1px solid #ddd;
        color:#555;
        font-size:14px;
        line-height:1.6;
    }
</style>
</head>

<body>

<div class="container">
    <h2>Palindrome Checker</h2>

    <input type="text" id="textInput" placeholder="Enter a word or
sentence">

    <button onclick="checkPalindrome()">Check</button>

    <p id="result"></p>

    <div class="footer">
        <p><strong>My Name:</strong> Manaswi Sakharwade</p>
        <p><strong>PRN:</strong> 24070521169</p>
    </div>
</div>

<script>
function checkPalindrome() {

    let input = document.getElementById("textInput").value;

    let text = input.toLowerCase().replace(/[^a-z0-9]/g, "");

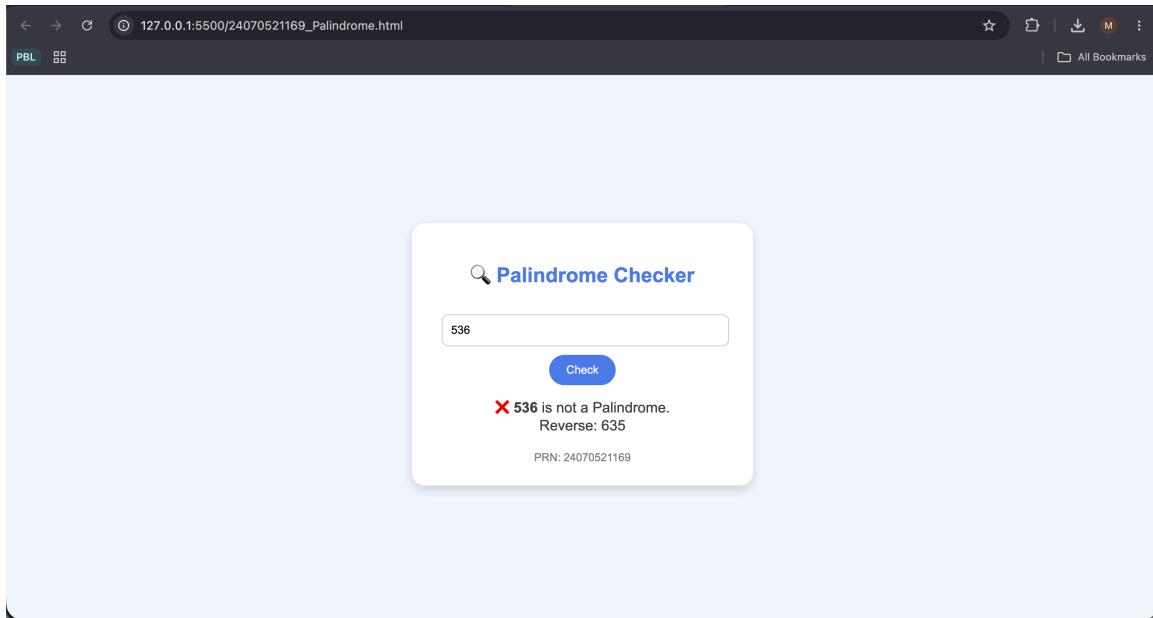
    let reverse = text.split("").reverse().join("");

    if(text === ""){
        document.getElementById("result").innerHTML = "Please enter
some text.";
    }
    else if(text === reverse){
        document.getElementById("result").innerHTML =
        ''' + input + ' is a <span style="color:green;">Palindrome
✔</span>';
    }
    else{
        document.getElementById("result").innerHTML =
        ''' + input + ' is <span style="color:red;">Not a Palindrome
✘</span>';
    }
}

```

```
}  
</script>  
  
</body>  
</html>
```

4. Output:



5. Case Study 04

Case Study Title:

Real-Time Case Study: ATM Card PIN Verification Using JavaScript Functions

Case Study Code:

- case_study4.html Code:

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
<meta charset="UTF-8">  
<meta name="viewport" content="width=device-width, initial-scale=1.0">  
<title>ATM PIN Verification</title>  
  
<style>  
body{  
    font-family: Arial, sans-serif;  
    background: linear-gradient(135deg, #dfe9f3, #ffffff);  
    display: flex;
```

```
        justify-content:center;
        align-items:center;
        height:100vh;
    }

    .container{
        width:400px;
        background:white;
        padding:25px;
        border-radius:12px;
        box-shadow:0 0 15px rgba(0,0,0,0.2);
        text-align:center;
    }

    h2{
        color:#0d47a1;
    }

    input{
        width:90%;
        padding:10px;
        margin:15px 0;
        font-size:18px;
    }

    button{
        padding:10px 18px;
        font-size:16px;
        background:#1565c0;
        color:white;
        border:none;
        border-radius:6px;
        cursor:pointer;
    }

    button:hover{
        background:#0d47a1;
    }

    #result{
        margin-top:20px;
        font-size:18px;
        font-weight:bold;
    }

    .footer{
        margin-top:25px;
        border-top:1px solid #ccc;
        padding-top:10px;
        font-size:15px;
    }
```

```
</style>
</head>

<body>

<div class="container">

<h2>ATM Card PIN Verification</h2>

<input type="password" id="pin" placeholder="Enter ATM PIN">

<br>

<button onclick="verifyPIN()">Verify PIN</button>

<div id="result"></div>

<div class="footer">
<b>Name:</b> Manaswi Sakharwade<br>
<b>PRN:</b> 24070521169
</div>

</div>

<script>

// Global Scope
let bankName = "ABC Bank";

// Function Declaration
function reversePIN(pin) {
    return pin.split("").reverse().join("");
}

// Function Expression
const checkPalindrome = function(pin) {
    return pin === reversePIN(pin);
};

// Arrow Function
const displayMessage = (msg) => {
    document.getElementById("result").innerHTML = msg;
};

function verifyPIN() {

    let pin = document.getElementById("pin").value;

    if(pin=="") {
        displayMessage("Please enter your ATM PIN.");
        return;
    }
}
```

```

    }

    let reversed = reversePIN(pin);

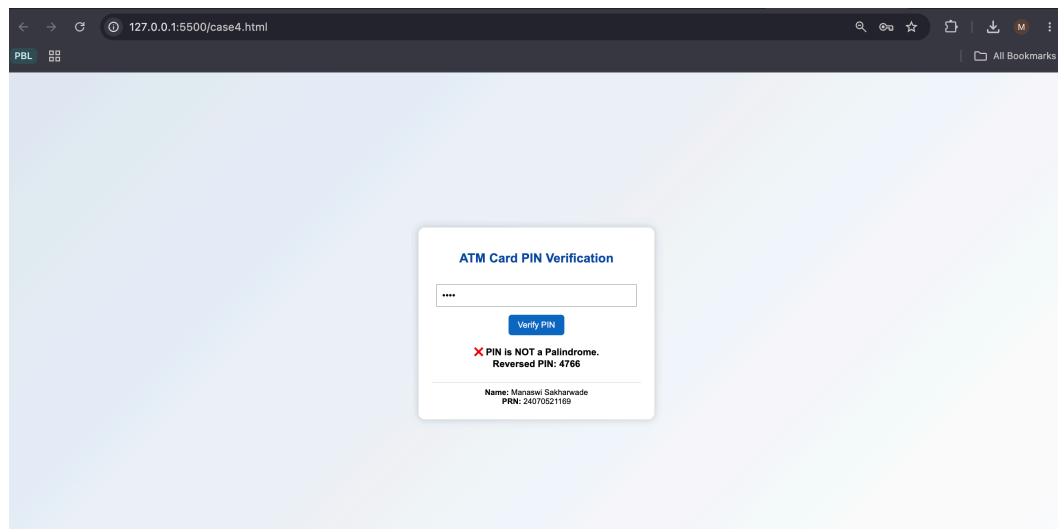
    if (checkPalindrome(pin)) {
        displayMessage(
            "✅ PIN is Palindrome.<br>" +
            "Reversed PIN: " + reversed +
            "<br>Special Security Message Displayed!"
        );
    }
    else{
        displayMessage(
            "❌ PIN is NOT Palindrome.<br>" +
            "Reversed PIN: " + reversed
        );
    }
}

</script>

</body>
</html>

```

Case Study Output:



Result/Conclusion:

The palindrome checker was successfully developed. A closure was effectively implemented to securely store the history of checked words. The try-catch mechanism successfully intercepted invalid inputs, preventing application crashes, and the core algorithm correctly identified whether the provided strings were palindromes.